

AWS HLS Video Streaming App - Cost Estimate Document

Project Overview

App Type: HLS Video Streaming Platform
Scenario: 1 hour video uploaded, watched by 10 users (one-time watch)
Currency: Indian Rupees (INR)

Key Features

- ✔ One IP/One Device login
- ✔ No screen recording (DRM protected)
- ✔ No casting allowed
- ✔ Multiple quality options (360p, 480p, 720p, 1080p)
- ✔ Speed control (0.5x to 2x)
- ✔ Forward/Backward seek

AWS Services Used

Service	Purpose
S3	Store video files and HLS segments
MediaConvert	Convert video to multiple qualities
CloudFront	Deliver video to users (CDN)
Lambda	Handle authentication & access control
RDS/DynamoDB	Store user data & device info
API Gateway	API endpoints for app

Video Specifications (1 Hour)

Quality	Resolution	File Size	Bitrate
360p	640×360	350 MB	0.8 Mbps

Quality	Resolution	File Size	Bitrate
480p	854×480	675 MB	1.5 Mbps
720p	1280×720	1.35 GB	3 Mbps
1080p	1920×1080	2.25 GB	5 Mbps

Total Storage Needed: ~5 GB (all qualities combined)

Cost Breakdown - One Time Watch (10 Users)

Step 1: Video Upload & Processing (One-time cost)

Item	Details	Cost (INR)
Video Transcoding	Convert to 4 qualities	₹4
Storage (S3)	5 GB stored for 1 month	₹9.20
Total Setup		₹13.20

Step 2: Video Delivery (When 10 users watch)

Scenario A: Everyone watches 100% video

Quality	Users	Data per User	Total Data	CDN Cost
1080p	10	2.25 GB	22.5 GB	₹153
720p	10	1.35 GB	13.5 GB	₹92
480p	10	675 MB	6.75 GB	₹46

Scenario B: Everyone watches 50% video

Quality	Users	Data per User	Total Data	CDN Cost
1080p	10	1.13 GB	11.3 GB	₹77
720p	10	675 MB	6.75 GB	₹46
480p	10	337 MB	3.37 GB	₹23

Scenario C: Mixed Quality (Most Realistic)

Quality	Users	Watch %	Data	Cost
1080p	2	100%	4.5 GB	₹31
720p	5	100%	6.75 GB	₹46
480p	3	50%	1 GB	₹7
Total	10	-	12.25 GB	₹84

Step 3: Security & Features Cost

Feature	Technology	Cost per Session
One IP/Device tracking	Lambda + DynamoDB	₹2
DRM (No recording/casting)	CloudFront Signed URLs	₹3
User Authentication	API Gateway + Lambda	₹1
Total Security		₹6

Final Cost Summary (10 Users, One-time Watch)

Option 1: Basic Quality (480p, 50% watch)

Video Setup	: ₹13.20
CDN Delivery	: ₹23
Security Features	: ₹6
TOTAL	: ₹42.20
Per User	: ₹4.22

Option 2: Good Quality (720p, 100% watch)

Video Setup	: ₹13.20
CDN Delivery	: ₹92
Security Features	: ₹6

TOTAL

: ₹111.20

Per User

: ₹11.12

Option 3: High Quality (1080p, 100% watch)

Video Setup

: ₹13.20

CDN Delivery

: ₹153

Security Features

: ₹6

TOTAL

: ₹172.20

Per User

: ₹17.22

Option 4: Mixed (Realistic Scenario)

Video Setup

: ₹13.20

CDN Delivery

: ₹84

Security Features

: ₹6

TOTAL

: ₹103.20

Per User

: ₹10.32

Monthly Projection (30 videos, 10 users each)

Scenario	Cost per Video	30 Videos/Month	Per User Cost
480p, 50% watch	₹42	₹1,260	₹4.20
720p, 100% watch	₹111	₹3,330	₹11.10
Mixed realistic	₹103	₹3,090	₹10.30

Cost Breakdown by Component (Monthly - 30 videos)

Component	Monthly Cost	% of Total
CDN (CloudFront)	₹2,520	82%
Storage (S3)	₹276	9%

Component	Monthly Cost	% of Total
Security (DRM)	₹180	6%
Transcoding	₹120	3%
TOTAL	₹3,096	100%

Ways to Reduce Costs

1. Optimize Video Quality

- Default to 720p instead of 1080p → Save 40% bandwidth
- Auto-detect user's internet speed

2. Enable CloudFront Caching

- First user pays full cost
- Next 9 users cost almost nothing
- **Savings:** Up to 70% on repeated views

3. Use S3 Lifecycle Policies

- Move old videos to cheaper storage
- **Savings:** 50% on storage after 30 days

4. Compress Videos Better

- Use H.265 instead of H.264
- **Savings:** 30-40% file size reduction

5. Region Selection

- Use Mumbai (ap-south-1) region
- Lower data transfer costs within India

Important Notes

1. **Free Tier Benefits** (First 12 months AWS):
 - 5 GB S3 storage free

- 1 million Lambda requests free
- Can save ₹500-800/month

2. Actual Costs May Vary:

- Based on actual user behavior
- Video compression efficiency
- Cache hit rates

3. Hidden Costs to Consider:

- API requests
 - Database reads/writes
 - Monitoring & logs
 - Add 10-15% buffer
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Recommended Starting Configuration

For Best Balance (Quality vs Cost):

- Default quality: 720p
- Allow switching to 480p/1080p
- Enable CloudFront caching
- Implement lazy loading
- Use Mumbai region

Expected Cost: ₹80-100 per video (10 users)

Per User: ₹8-10

Next Steps

1. Set up AWS account
 2. Configure S3 bucket
 3. Set up MediaConvert for transcoding
 4. Configure CloudFront distribution
 5. Implement Lambda functions for security
 6. Build authentication system
 7. Test with sample video
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Document prepared for: HLS Video Streaming App

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All costs are approximate and based on AWS Mumbai region pricing